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RESEARCH ARTICLE

Heterogeneous Committee-Based Adaptive Active Learning for Efficient Heart Disease Prediction

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ABSTRACT Heart disease remains the leading cause of mortality worldwide. Early diagnosis is essential, yet traditional machine learning-based diagnostic systems often rely heavily on large amounts of expert-labeled data, making them costly and time-consuming to develop. To address this challenge, we propose a heterogeneous committee-based adaptive active learning (HC-AAL) framework for heart disease prediction. The method integrates multiple machine learning techniques, including XGBoost, Random Forest, and Logistic Regression, into a diverse committee that iteratively selects the most informative samples using a committee-weighted query-by-committee (CW-QBC) uncertainty measure. To enhance diversity and representativeness, K-Means clustering is applied to uncertain instances, and adaptive query control is used to dynamically adjust the number of queried samples per iteration. In particular, the adaptive query length mechanism dynamically adjusts the number of samples queried at each iteration to balance learning efficiency and computational cost. Experimental results demonstrate that the proposed HC-AAL framework achieves 96.04% accuracy using only 31.9% of labeled data, significantly outperforming traditional fully supervised approaches. These findings highlight the effectiveness of adaptive active learning strategies in reducing annotation costs while maintaining high predictive performance.

INDEX TERMS Committee-based adaptive learning, heart disease prediction, machine learning.

I. INTRODUCTION

Heart disease refers to any condition that affects the normal functioning of the heart. It is the primary cause of death worldwide, contributing to a large percentage of annual global fatalities [1]. In China, the number of heart disease–related deaths increased from 277,312 in 2013 to 463,314 in 2021, accounting for over 25% of all deaths nationwide [2]. In 2024, heart disease accounted for nearly 20.5 million deaths in India [3]. Heart disease is also highly prevalent in the United States, where the American Heart Association estimates that around 82.6 million individuals are affected [4]. Obesity, high cholesterol, alcohol use, lack of physical activity, high blood pressure, diabetes, and stress are among the main contributors to heart disease. In the absence of effective prevention, heart disease often leads to death [5]. The use of machine learning in medical diagnosis is rapidly increasing, as it can reduce human errors and improve diagnostic accuracy [6]. Several studies have explored the use of machine learning models for heart disease diagnosis, focusing on both classification and prediction tasks. In a detailed analysis, [7] explored the role of machine learning in the detection of heart disease. According to this review, existing studies on heart disease detection can be broadly classified into three categories: (i) clinical information–based detection, (ii) ECG and PCG signal–based detection, and (iii) X-ray image–based detection. In this work, we also focus on clinical information-based evidence. According to [7], studies utilizing standard clinical information, advanced models such as XGBoost, Random Forest, Ensemble Learning, and Neural Networks outperform conventional machine learning techniques. For instance, [8] employed six machine learning methods, including XGBoost, Bagging, Random Forest, Decision Tree, K-Nearest Neighbor, and Naïve Bayes. Among these, XGBoost achieved the best performance with an accuracy of 91.3%. In another study, [9] employed a hybrid

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machine learning approach (Decision Tree + AdaBoost) to predict and classify heart disease. Model performance is evaluated using accuracy, True Positive Rate, and specificity. However, most of the previous studies utilize 70–80% of labeled data to develop machine learning models. Despite the growing availability of data, collecting labeled data remains challenging as it demands significant effort from domain experts and specialists for accurate annotation [10]. This challenge is particularly important in supervised learning, which relies on labeled data for model training. Since labeling samples requires expert input, it is costly and time-consuming. To mitigate this, prioritizing informative samples is essential. Active Learning (AL) is a technique designed to identify the most informative samples, which is the focus of the proposed approach.

Active Learning (AL) methods employ supervised machine learning iteratively, where the most informative samples are automatically selected for expert annotation. This approach effectively keeps humans in the loop while significantly reducing the amount of annotated data needed compared with traditional supervised ML methods. Several query selection strategies have been developed to identify the most informative unlabeled instances for annotation. The most commonly used approaches include Uncertainty Sampling, where the learner selects samples for which the model is least confident; Query-by-Committee (QBC), where a committee of models trained on the labeled set votes on candidate instances and the sample with the highest disagreement is chosen; Expected Model Change, a decision-theoretic strategy that selects samples expected to induce the largest change in the current model; Expected Error Reduction, which targets instances likely to reduce future prediction error; Variance Reduction, which aims to minimize prediction uncertainty; and Density-Weighted Methods, which combine informativeness with representativeness to avoid selecting outliers [11]. However, the use of AL in clinical information extraction remains limited, particularly for medical concept identification [10]. Unlike these existing AL strategies that rely on a single criterion, such as uncertainty, committee disagreement, or density weighting, the proposed method introduces a hybrid multi-criterion framework.

In this work, we propose a heterogeneous committee-based adaptive active learning (HC-AAL) approach for heart disease prediction using the publicly available Heart Disease Dataset [12], which contains clinical and diagnostic data from four sources: Cleveland, Hungary, Switzerland, and Long Beach V. The dataset includes 76 attributes, with 14 key features—such as age, sex, chest pain type, blood pressure, cholesterol, ECG results, and exercise test outcomes—commonly used for prediction. The proposed framework integrates the strengths of multiple machine learning models, including XGBoost, Random Forest, and Logistic Regression, to form a diverse committee. At each iteration, the committee collaboratively evaluates the unlabeled data and selects the most informative samples based on a committee-weighted query-by-committee (CW-QBC) uncertainty measure. To further enhance diversity among the selected samples, we apply K-Means clustering to the uncertain instances and select representative points closest to each cluster centroid. This adaptive querying process is one of the key innovations of the proposed approach since it dynamically adjusts the query size across iterations, ensuring a balance between learning efficiency and computational cost. In addition, the iterative selection strategy enables efficient learning that boosts high performance with minimal labeling effort. Compared with traditional AL methods, this approach advances sample selection by simultaneously incorporating informativeness, diversity, and adaptive control of labeling effort. Experimental results on the heart disease dataset show that the proposed HC-AAL method achieves 96.04% accuracy using only 31.9% of labeled data, significantly outperforming traditional fully supervised learning approaches. The main contributions of this paper are summarized as follows:

- We propose a novel heterogeneous committee-based adaptive active learning (HC-AAL) framework for heart disease prediction.
- We develop the CW-QBC uncertainty estimation method to improve query selection efficiency.
- We incorporate diversity-based K-Means sampling and adaptive query control to enhance sample representativeness.
- Our approach stands out from existing AL methods by integrating uncertainty, committee disagreement, diversity sampling, and adaptive querying into a single hybrid framework.
- In summary, we introduce a novel heterogeneous committee-based adaptive active learning (HC-AAL) framework that unifies uncertainty estimation, committee disagreement, diversity-driven sampling, and adaptive query control to significantly enhance heart disease prediction with minimal labeled data.
- We validate the proposed framework on the heart disease dataset, which contains clinical and diagnostic data from four sources: Cleveland, Hungary, Switzerland, and Long Beach V [12], achieving superior performance with significantly reduced labeling cost.

The remainder of this paper is organized as follows. Section II presents the related Literature Review. Section III describes the details of the Proposed Work. Section IV discusses the Experimental Analysis and results. Finally, Section V provides the Conclusion and possible directions for future work.

II. LITERATURE REVIEW

Heart disease refers to conditions that affect the structure and function of the heart. Early detection is crucial to reducing the risk of serious complications, such as heart attacks and heart failure. Machine learning models can analyze historical patient data to identify risk factors and support the timely and reliable detection of heart disease [13]. Machine learning approaches for heart disease prediction have progressed rapidly, demonstrating the successful combination of medical science and technology [13]. Numerous machine learning methods have been applied to heart disease prediction and can be broadly grouped into two categories: traditional ML-based techniques and active learning–based approaches.

A. TRADITIONAL ML-BASED TECHNIQUES

Traditional ML models are widely used in heart disease prediction, leveraging clinical attributes for risk classification and depending on fully labeled data for supervised training. For instance, [14] designed an Effective Heart Disease Prediction System (EHDPS) that integrates multiple machine learning algorithms (K-Nearest Neighbors (KNN), Logistic Regression, and Random Forest) to identify individuals at risk of cardiovascular disease. The system was evaluated using the UCI Heart Disease dataset, including 303 instances and 14 medical attributes. Data normalization and missing value imputation were applied to improve model consistency. Among the models, KNN achieved the highest classification accuracy of 88.52%. In another study, [15] utilized the Cleveland Heart Disease Dataset obtained from the UCI repository, which comprised 12 key features after data cleaning. They evaluated Logistic Regression, Random Forest, and Support Vector Machine algorithms. The results revealed that Random Forest achieved the highest accuracy (72.33%), while Logistic Regression outperformed the others in terms of ROC-AUC (0.7810). To improve heart disease prediction, [5] proposed a hybrid model combining a Deep Neural Network (DNN) for feature extraction, Power Transform for normalization, Principal Component Analysis (PCA) for dimensionality reduction, and Logistic Regression (LR) for classification. They used the Cleveland Heart Disease Dataset from the UCI repository, containing 297 instances with 13 selected features. With a 70–30 train-test split, the model achieved 93.33% accuracy and demonstrated superior performance compared to existing approaches. [16] analyzed patient data from Khyber Teaching Hospital and Lady Reading Hospital, Pakistan, for heart disease prediction. Decision Tree, Random Forest, Logistic Regression, Naïve Bayes, and SVM were used in this study. Model evaluation using exploratory data analysis, confusion matrices, and ROC analysis showed that Random Forest performed best, achieving an accuracy of 85.01%. [17] Applied machine learning to predict heart disease from patient health data. They tested MLP, Random Forest, Naïve Bayes, and SVM models with preprocessing and feature selection. The SVM model achieved an accuracy of 91.67%. [18] developed a hybrid ensemble approach that integrates several machine learning algorithms to predict heart disease. It combined SVM, gradient boosting, decision tree, logistic regression, KNN, and random forest. The model achieves 92.95% accuracy and precision. A hybrid model was proposed by [19] for predicting cardiovascular disease (CVD), which combines Random Forests to capture complex nonlinear relationships with mixed-effect components from GLMM to account for clustered data. The model achieved an accuracy of 0.75 and an AUC of 0.80, outperforming standard Decision Tree, Random Forest, and GLMM Tree models. In the work of [20], a hybrid XGBoost classifier with hyperparameters optimized using the Optuna framework was combined with outlier removal techniques (z-score and IQR). The highest accuracy of 95.45% was achieved on the Cleveland HD dataset when outliers were removed and a 90:10 train-test split was applied. [21] developed a machine learning framework to predict cardiovascular disease (CVD). Six machine learning models (DT, LR, NB, SVM, RF, and XGBoost) were applied to SMOTE-balanced data. The models were trained using 10-fold cross-validation, with 80% of the data used for training, and hyperparameters optimized through grid and randomized search. Two experiments compared predictions with and without geographical features. XGBoost achieved the highest accuracy of 95.24%, highlighting the importance of including geographical factors. The proposed work by [22] presents a hybrid approach for predicting heart disease, integrating six machine learning classifiers (Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Support Vector Machine, and XGBoost) with a deep learning sequential model. Performance is evaluated on the SMOTE-balanced UCI Heart Disease dataset through accuracy and confusion matrices. The deep learning model achieved the highest accuracy of 94.2%. [6] developed an advanced machine learning-based approach for heart disease prediction utilizing the Cleveland dataset, where 70% of the data is allocated for training. The model combines Random Forest and Decision Tree methods to achieve higher accuracy and real-time diagnostic support. The proposed hybrid model achieves an accuracy of 88.7%, surpassing individual classifiers. A machine learning framework for heart disease prediction was proposed by [23], employing Random Forest, K-Nearest Neighbors (KNN), and Decision Tree classifiers. The data were preprocessed, categorical variables were encoded, and features were scaled before training. Ten-fold cross-validation was applied, with 90% of the dataset used for training in each fold, and patient clinical features were considered for model evaluation. Among the classifiers, Random Forest achieved the best performance, with an accuracy of approximately 83%. The Classification and Regression Tree (CART) algorithm, as applied by [24], was employed to predict heart disease from the Aggregated dataset, including 11 clinical variables. The model was trained on 80% of the data, with decision rules derived from both categorical and numerical variables. It achieved 87.25% accuracy. According to [25], a CatBoost-based machine learning framework was proposed for heart disease prediction. The dataset used in the model combines five UCI Heart Disease datasets into a single collection comprising 918 records and 11 features. Data preprocessing addressed missing values and applied SHAP for feature selection. After comparing multiple algorithms (XGBoost, AdaBoost, Random Forest, and SVC), the CatBoost model outperformed the others. Hyperparameter optimization using Optuna resulted in an accuracy of 90.94%. [26] proposed an XAI-based ensemble classification framework for heart disease prediction using a cardiovascular dataset containing 303 samples and 14 features. The model integrates SVM, AdaBoost, KNN, Bagging, Logistic Regression, and Naïve Bayes within an explainable AI environment. Their XAI-based approach achieved a notable accuracy of 89%, outperforming conventional models. In all the methods discussed above, 70% or more of the data is used for training to achieve the reported outcomes.

B. ACTIVE LEARNING-BASED APPROACHES

Unlike traditional ML models that depend on fully labeled datasets, active learning aims to reduce labeling cost by selectively querying the most informative samples. This makes it highly suitable for medical domains, where expert annotation is expensive and time-consuming. As highlighted by [27], the most heart disease prediction studies rely on memorization-based methods that capture relationships between dataset features and labels. Active learning approaches for improving generalization in heart disease diagnosis remain largely unexplored. They implemented an active learning framework for multi-label heart disease prediction using five selection methods (MMC, Random, Adaptive, QUIRE, and AUDI) combined with a hyperparameter-tuned label-ranking model. The Adaptive method achieved the highest performance, with an accuracy of 57.4% when the labeled sample cost was at least 40. In another study, [4] proposed a probabilistic KNN enhanced with MCMC-based metric learning and active learning for heart disease prediction. The model uses a dataset of 299 samples with 13 features, split into $\approx$ 70% labeled for training and $\approx$ 30% unlabeled/test, and iteratively queries the most uncertain instances to improve learning. Using $k = 5$ nearest neighbors (number of neighbors considered for classification) and $Q = 10$ queries (number of most uncertain instances added per iteration), it achieves 92.22% accuracy and 81.08% F1-score, outperforming baseline classifiers such as RF, SVM, and ANN. In this work, we introduce a novel active learning framework for heart disease prediction, designed to achieve high performance using minimal labeled data. Initially trained on only 1.5% of labeled data, the model incrementally improves by querying the most informative samples. Uncertainty is quantified using the Committee-Weighted Query-by-Committee (CW-QBC) strategy, which integrates the outputs of a diverse ensemble comprising XGBoost, Random Forest, and Logistic Regression models. To ensure diversity among the queried samples, a K-Means–based sampling strategy is applied to select points closest to the cluster centers. The model achieves 96.04% accuracy while using only 31.9% of the dataset for training, demonstrating superior performance with limited labeled data.

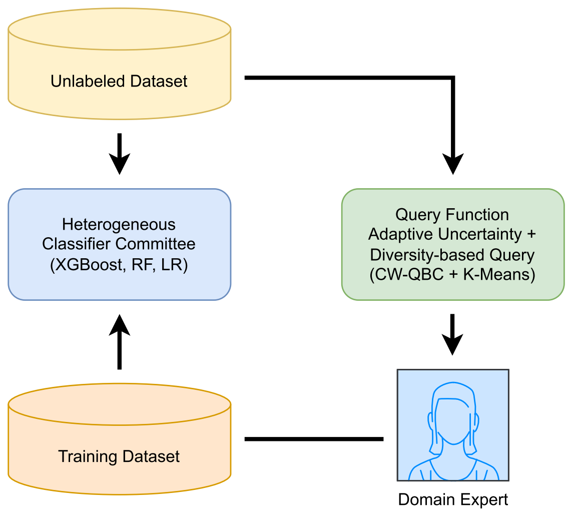

FIGURE 1. Overview of the proposed heterogeneous committee-based AL framework using the hybrid multi-criterion.

III. PROPOSED WORK

In this work, a novel active learning (AL) framework is introduced for heart disease classification, integrating both uncertainty- and diversity-based sampling criteria. The proposed framework employs a heterogeneous committee of classifiers, including XGBoost, Random Forest, and Logistic Regression, to evaluate unlabeled samples. Sample uncertainty is measured using a Committee-Weighted Query-by-Committee (CW-QBC) approach, which quantifies disagreement among the models to identify the most informative instances. Furthermore, K-Means clustering is applied to the most uncertain samples to ensure the selection of diverse and representative data points in the feature space. This integration enables the framework to balance accuracy and data efficiency, providing a practical and cost-effective solution for heart disease diagnosis. Figure 1 shows an overview of the proposed general model.

A. INITIAL LABELED SET

Active Learning (AL) enables models to learn effectively from fewer labeled examples by actively selecting which data to label [11]. Since labeling data often requires costly input from human experts, it is important to build an accurate prediction model using only a small number of labeled examples [27]. Following the standard practice in AL literature [11], we initialize the labeled training set with a small fraction of the dataset to simulate a real-world scenario where labeling is expensive. In our experiments, we select 1.5% of the dataset as the initial labeled set. This fraction was chosen as a design decision based on an ablation study that evaluated different initial fractions and found that 1.5% provides a good trade-off between initial model performance and labeling efficiency.

Algorithm 1 Initial Labeled Set Selection

Require: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, initial fraction ρ (e.g., $\rho = 0.015$), random seed s
Ensure: Initial labeled set $\mathcal{L}_0$, Unlabeled pool $\mathcal{U}_0$

// Split features and labels
$X \leftarrow \{x_i\}_{i=1}^{N}, \quad Y \leftarrow \{y_i\}_{i=1}^{N}$
// Compute initial labeled set size
$n_0 \leftarrow \max\left(1, \lfloor \rho \cdot N \rfloor\right)$ $\triangleright$ ensure at least one sample
// Reproducible random partition: equivalent to `train_test_split(..., train_size=` ρ`, random_state=`s`)`
Randomly permute indices $I = \text{permute}(1, \ldots, N; \textit{seed} = s)$
$I_{\text{train}} \leftarrow I[1 : n_0]$
$I_{\text{left}} \leftarrow I[n_0 + 1 : N]$
// Build labeled and unlabeled sets
$\mathcal{L}_0 \leftarrow \{(X[i], Y[i]) \mid i \in I_{\text{train}}\}$
$\mathcal{U}_0 \leftarrow \{(X[i], Y[i]) \mid i \in I_{\text{left}}\}$
return $\mathcal{L}_0, \mathcal{U}_0$

The dataset is defined as:

$$D = \{(x_i, y_i)\}_{i=1}^{N}, \tag{1}$$

where x_i is the feature vector of the i-th instance, and y_i is its label.

We start by randomly selecting a small portion of the data as the *initial labeled set*:

$$D_{\text{labeled}}^{(0)} \subset D, \quad |D_{\text{labeled}}^{(0)}| = 0.015 \cdot N. \tag{2}$$

The remaining data forms the *unlabeled pool*:

$$D_{\text{unlabeled}} = D \setminus D_{\text{labeled}}^{(0)}. \tag{3}$$

This method represents real situations where labeling data is costly. The active learning algorithm selects the most informative samples from $D_{\text{unlabeled}}$ to efficiently improve the model. Algorithm 1 summarizes the initialization and iterative selection procedure used in the proposed active learning framework.

B. HETEROGENEOUS COMMITTEE DESIGN

In contrast to homogeneous ensembles that rely on models with identical structures (e.g., multiple instances of the same learning algorithm trained on different data subsets), heterogeneous ensembles integrate classifiers from diverse algorithmic families [28]. This diversity allows the ensemble to capture multiple perspectives of the underlying data distribution. When their complementary biases are combined, the ensemble approach effectively mitigates individual model weaknesses and strikes a balance between bias and variance. As a result, it achieves greater accuracy and robustness compared to homogeneous ensembles [29]. As shown in Algorithm 2, the proposed Active Learning (AL) framework in this work employs a heterogeneous committee of classifiers. The committee consists of three classifiers based on distinct learning algorithms (XGBoost, Random Forest, and Logistic Regression) to achieve a balanced trade-off between accuracy, robustness, and interpretability.

The XGBoost classifier is integrated as a key component to handle the binary prediction task. Its design follows a boosting strategy, where multiple decision trees are trained sequentially, and each new tree focuses on correcting the errors made by the previous ones. This iterative learning process allows the model to capture complex, non-linear relationships among clinical features. Within the framework, XGBoost is used to provide a strong and reliable learner that complements other classifiers in the committee. During the active learning process, it is repeatedly retrained with the newly labeled data to evaluate how model performance improves as more information becomes available. At the end of the iterative process, XGBoost serves as the final classifier to perform the overall evaluation of the system. Ensemble diversity is strengthened by the Random Forest classifier through the development of multiple decision trees, each derived from randomized selections of the data and feature space. This randomness reduces model variance and prevents overfitting, leading to more stable and generalized predictions. The ensemble voting process of Random Forest mitigates the influence of noisy features, and its proficiency in representing non-linear feature interactions complements the gradient-based optimization used in XGBoost. Within the committee, Random Forest contributes alternative probabilistic estimates that strengthen uncertainty measurement and improve the reliability of sample selection. Logistic Regression functions as an interpretable and computationally efficient model that complements the complexity of tree-based classifiers. The model predicts the probability of each class using a logistic function, which results in a linear boundary between the classes. Implemented as part of a standard preprocessing pipeline, it preserves numerical stability and normalizes the influence of all features. Despite its simpler structure, Logistic Regression introduces a distinct algorithmic perspective by modeling linear relationships that tree-based classifiers may overlook. Its complementary contribution supports a more balanced and robust interpretation of the data distribution.

C. ADAPTIVE UNCERTAINTY AND DIVERSITY-BASED QUERY DESIGN

To efficiently expand the labeled dataset, the proposed active learning framework employs an adaptive uncertainty and diversity-based query strategy. This strategy leverages the heterogeneous committee of classifiers described earlier to identify the most informative samples from the unlabeled pool. In each iteration, all committee members (XGBoost, Random Forest, and Logistic Regression) predict the class probabilities of the unlabeled samples. The Committee-Weighted Query-by-Committee (CW-QBC) approach is then applied to quantify the level of disagreement among the models. [30], [31].

Algorithm 2 Heterogeneous Committee Design

Require: Labeled dataset $\mathcal{L} = \{(x_i, y_i)\}_{i=1}^{n}$, base learners $\{M_1, M_2, \ldots, M_k\}$
Ensure: Trained heterogeneous committee

$$\mathcal{C} = \{M_1, M_2, \ldots, M_k\}$$

Define base learners:
- M_1: XGBoost classifier (`XGBClassifier`)
- M_2: Random Forest classifier
- (`RandomForestClassifier`)
- M_3: Logistic Regression with data standardization pipeline

for each model $M_j \in \{M_1, M_2, M_3\}$ **do**
  Train M_j on $\mathcal{L}$ using features x_i and labels y_i
end for
Form the heterogeneous committee:

$$\mathcal{C} = \{M_1, M_2, M_3\}$$

return $\mathcal{C}$

CW-QBC extends entropy-based QBC disagreement by operating on probabilistic predictions and enabling confidence-weighted contributions from heterogeneous committee members. Suppose M denotes the number of models in the committee, and consider $p_m(x_i)$ represents the probability predicted by model m for sample x_i belonging to the positive class. The average probability across the committee is computed as:

$$\bar{p}(x_i) = \frac{1}{M} \sum_{m=1}^{M} p_m(x_i) \tag{4}$$

The CW-QBC score for each sample x_i is then calculated as:

$$CW\_score(x_i) = \sum_{m=1}^{M} p_m(x_i) \cdot \log \frac{p_m(x_i) + \epsilon}{\bar{p}(x_i) + \epsilon} \tag{5}$$

where ϵ is a small constant added to prevent division by zero. Let Q denote the set of top uncertain samples and k the number of samples to query. K-Means clusters Q into k groups:

$$\{C_1, C_2, \ldots, C_k\} = \text{KMeans}(Q, k) \tag{6}$$

For each cluster C_j, the sample closest to the centroid μ_j is selected:

$$x_j^* = \arg\min_{x \in C_j} \|x - \mu_j\|_2 \tag{7}$$

Algorithm 3 provides a summary of the adaptive query selection process. In summary, after creating the initial labeled set, the framework expands it iteratively. At each iteration, the heterogeneous committee is retrained on the current labeled set, and CW-QBC scores are computed to identify the most uncertain samples. Based on an adaptive query size, the algorithm chooses the most uncertain samples, clusters them with K-Means, and selects one representative per cluster to ensure diversity. The selected samples are then added to the labeled set and removed from the unlabeled pool. Model performance is continuously monitored on a hold-out test set, and the process repeats until a stopping criterion, such as reaching a maximum number of iterations or sufficient labeled data, is met. In a real-world active learning scenario, the selected samples would be sent to a domain expert for annotation before being added to the labeled dataset. However, in this work, to ensure a controlled and reproducible evaluation, the expert annotation process is simulated using the known ground-truth labels of the selected instances. In the experimental implementation, this iterative process was executed for 78 iterations to ensure convergence and to assess the highest performance across successive labeling stages.

Algorithm 3 Adaptive Uncertainty and Diversity-Based Query

Require: Unlabeled pool $\mathcal{U}$, initial labeled set $(\mathcal{X}_L, \mathcal{Y}_L)$, committee of M classifiers $\{C_m\}_{m=1}^{M}$, adaptive query size schedule $\mathcal{Q} = \{q_t\}$, maximum iterations T
Ensure: Expanded labeled set $(\mathcal{X}_L, \mathcal{Y}_L)$

Initialize iteration $t \leftarrow 0$
while $t < T$ **and** stopping criterion not met **do**
  Train each classifier C_m on $(\mathcal{X}_L, \mathcal{Y}_L)$
  for each sample $x \in \mathcal{U}$ **do**
    Compute predicted probabilities $p_m(x)$ from each C_m
    Compute mean probability $\bar{p}(x) = \frac{1}{M} \sum_{m=1}^{M} p_m(x)$
    Compute CW-QBC uncertainty score:

$$s_{\text{CW}}(x) = \sum_{m=1}^{M} p_m(x) \log\left(\frac{p_m(x) + \varepsilon}{\bar{p}(x) + \varepsilon}\right)$$

  end for
  Set adaptive query size $q \leftarrow \mathcal{Q}[t]$
  Select top $2q$ uncertain samples $\mathcal{U}_{\text{uncertain}}$ with highest s_{CW}
  Apply K-Means clustering $(k = q)$ on $\mathcal{U}_{\text{uncertain}}$
  For each cluster, select sample closest to centroid
  Query oracle (or reveal ground-truth) for labels of selected samples
  Update labeled set:

$$\mathcal{X}_L \leftarrow \mathcal{X}_L \cup \{x_j\}_{j=1}^{q}, \quad \mathcal{Y}_L \leftarrow \mathcal{Y}_L \cup \{y_j\}_{j=1}^{q}$$

  Remove selected samples from $\mathcal{U}$
  $t \leftarrow t + 1$
end while
return $(\mathcal{X}_L, \mathcal{Y}_L)$

IV. EXPERIMENTAL ANALYSIS

A. DATASET

The proposed method is evaluated using the Heart Disease dataset, which was collected from four sources: Cleveland, Hungary, Switzerland, and Long Beach V, and is widely

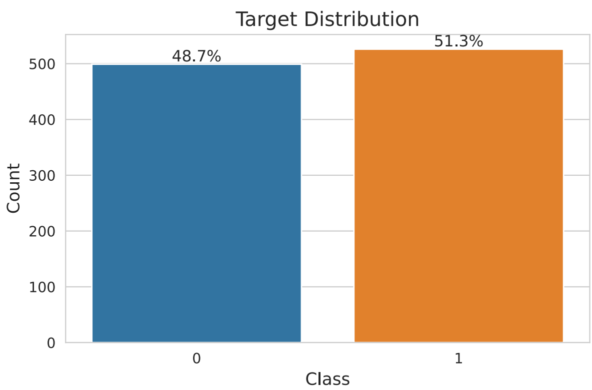

FIGURE 2. Target class distribution.

TABLE 1. Heart disease dataset attributes.

Attribute	Description
age	Patient age
sex	Gender
cp	Chest pain type
trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar >120 mg/dl
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	ST depression induced by exercise
slope	Slope of peak exercise ST segment
ca	Number of major vessels
thal	Thalassemia
target	Presence of heart disease

available on the Kaggle platform for machine learning datasets [12]. The dataset contains 1,025 instances with 14 attributes, including the target variable. The target variable, heart_disease, indicates whether a patient has heart disease (0 = no, 1 = yes). The dataset is approximately balanced, with 526 positive cases and 499 negative cases, eliminating the need for any balancing techniques.Figure 2 shows the distribution of the target classes in our dataset. Furthermore, there are no missing values in any of the attributes. All features are numeric, with most represented as integers (int64) and one continuous feature, oldpeak, represented as a float (float64). Table 1 summarizes the attributes of the Heart Disease dataset.

B. EVALUATION METRICS

To evaluate the proposed approach, this study uses several standard metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, which provide a detailed understanding of model performance. A description of each metric [32] is provided below:

Accuracy measures the overall correctness of the model predictions and is calculated as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \tag{8}$$

Precision indicates the proportion of true positive predictions among all instances classified as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{9}$$

Recall measures the proportion of true positive predictions among all actual positive instances:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{10}$$

F1-score is the harmonic mean of precision and recall, providing a balance between the two:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

AUC-ROC quantifies the ability of the model to distinguish between classes and is obtained from the area under the ROC curve.

C. EXPERIMENTAL SETUP

All computational experiments were executed in Python using standard machine learning and data analysis libraries. We evaluate our proposed active learning framework using a heterogeneous committee of classifiers: XGBoost, Random Forest, and Logistic Regression. StandardScaler is applied only within the Logistic Regression pipeline since tree-based models are scale-invariant and do not require feature normalization [33]. K-Means clustering is used to select diverse samples from the most uncertain points in each iteration.

The active learning setup uses an initial labeled set of 15 samples (1.5% of the 1,025-instance dataset) and an initial unlabeled pool of 1,010 samples. In each active learning iteration, 2 × query_size samples (ranging from 6 to 10) are first selected based on uncertainty, and query_size samples (ranging from 3 to 5) are finally selected using K-Means diversity. Each experiment is repeated 10 times to account for variability, and the active learning loop runs for 78 iterations per experiment, tracking performance metrics at each step.

All hyperparameters used in the experimental setup are summarized in Table 2.

D. RESULT EXPLANATION

The proposed active learning framework was evaluated on the heart disease dataset using a heterogeneous committee consisting of XGBoost, Random Forest, and Logistic Regression models. The system began with only 1.5% of the labeled data and iteratively queried the most informative samples based on committee disagreement (using the CW-QBC measure) and K-Means clustering to ensure diversity among the queried instances. To obtain statistically consistent results and reduce the influence of randomness, the experiment was conducted 10 times, and the reported metrics represent the mean values over these runs, shown in Figure 3.

Using only 31.9% of the dataset for training, the model achieved an accuracy of 96.04%, AUC of 97.08%, precision of 95.24%, recall of 98.07%, and an F1-score of 96.62%. The high accuracy (96.04%) and AUC (97.08%) indicate that the

TABLE 2. Hyperparameters of the active learning framework.

Component	Hyperparameter	Value / Description
Logistic Regression	max_iter	1000
Logistic Regression	random_state	42
Random Forest	n_estimators	100
Random Forest	random_state	42
XGBoost	objective	binary:logistic
XGBoost	use_label_encoder	False
XGBoost	eval_metric	logloss
XGBoost	random_state	42
KMeans	n_clusters	Adaptive query size (3–5)
KMeans	random_state	42
KMeans	n_init	10
Active Learning	initial_train_fraction	1.5% of total data
Active Learning	query_sizes	3–5 (adaptive)
Active Learning	iterations	78
Active Learning	num_runs	10

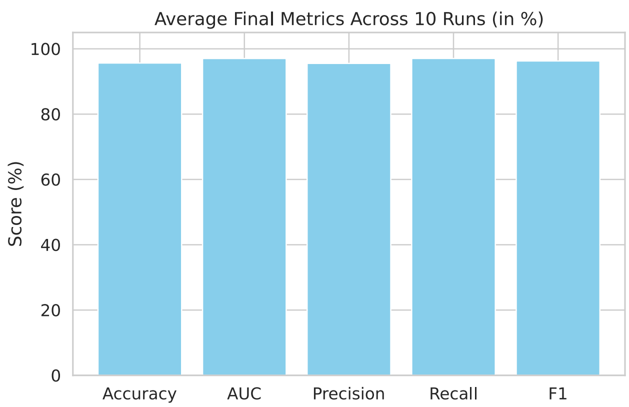

FIGURE 3. Average performance metrics (accuracy, AUC, precision, recall, and F1-score) of the proposed framework over multiple runs.

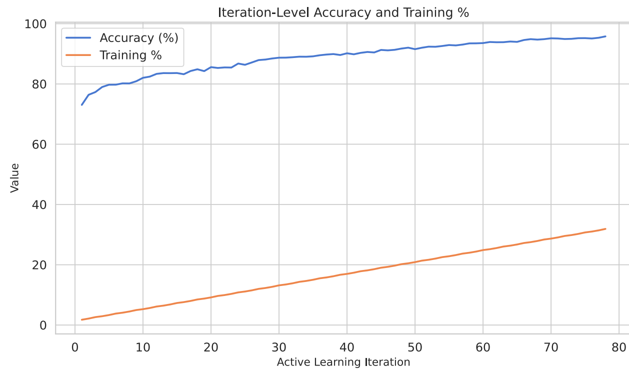

FIGURE 4. Performance metrics across active learning iterations.

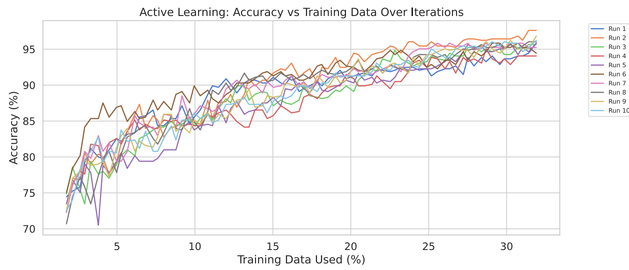

FIGURE 5. Model accuracy across active learning iterations using different amounts of training data.

proposed model effectively distinguishes between patients with and without heart disease. The high recall (98.07%) shows that the model successfully identifies nearly all positive (disease) cases, which is critical in medical diagnosis to minimize false negatives. The precision (95.24%) confirms that most of the positive predictions are correct, meaning few false positives. The F1-score (96.62%) balances precision and recall, indicating strong overall classification performance. Figure 4 illustrates the evolution of model accuracy and the proportion of labeled data across successive active learning iterations. As shown in the figure, a noticeable increase in accuracy is observed during the initial iterations with minimal labeled data. The continuous improvement in accuracy confirms that the proposed framework efficiently selects informative samples, achieving near-optimal performance (96%) with only 31.9% labeled data.

To ensure the reliability of the reported performance, the active learning process was repeated 10 times. Figure 5 presents the accuracy trends across these runs. Minor fluctuations from random sampling are observed; however, accuracy in all runs follows a consistent upward trend, demonstrating the robustness and stability of the proposed framework.

To further validate the consistency between different evaluation metrics, the relationship between accuracy and AUC across 10 runs is illustrated in Figure 6. Each data point corresponds to one run of the experiment, indicating small fluctuations in accuracy and AUC across repeated trials. As depicted, the results remain consistently high, with accuracy ranging between 95% and 97% and AUC between 96% and 98%. The strong positive correlation between the two metrics indicates that the proposed framework achieves stable and reliable classification performance across different runs, confirming its robustness against random initialization and data sampling effects.

E. EFFECT OF ADAPTIVE QUERY SIZE

The proposed model employs an adaptive query size strategy, dynamically adjusting the number of queried samples within the range of 3 to 6 (query_sizes = list(range(3, 6))). To evaluate its effectiveness, we compared the adaptive strategy against fixed query sizes of 3, 4, 5, 8, and 10. All results reported in Table 3 represent the average performance over 10 runs. As the table depicts, the adaptive query size approach achieves a strong balance between performance and labeling efficiency. Its Accuracy (96.04%), Recall (98.07%), and F1-Score (96.62%) surpass all fixed query sizes. While fixed sizes of 3, 4, and 5 are within the adaptive range,

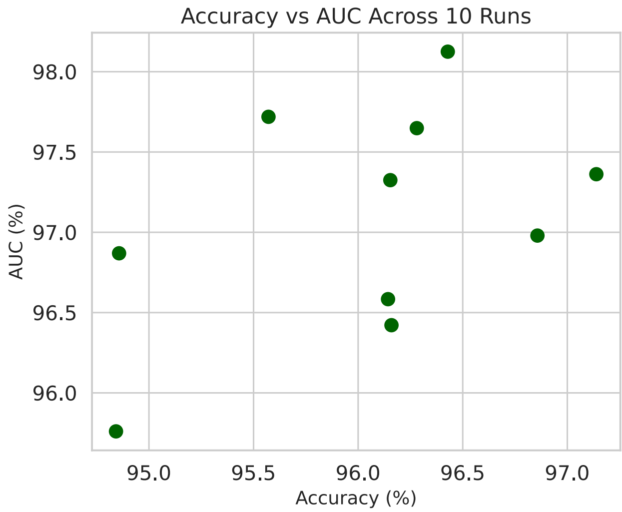

FIGURE 6. Relationship between accuracy and AUC across 10 independent runs, demonstrating consistent and reliable model performance.

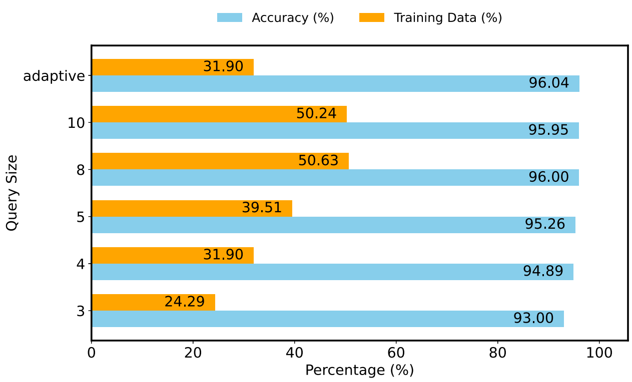

FIGURE 7. Accuracy vs. Training data for fixed and adaptive query sizes.

dynamically adjusting the query size ensures more consistent and effective learning since it dynamically selects the number of samples based on the current uncertainty and informativeness of the model. The small size of 3 can slow convergence by providing too few informative samples per iteration. The adaptive approach effectively balances exploration and exploitation, selecting smaller batches when fine-grained updates are needed and larger batches when more information can accelerate learning. Although fixed query sizes of 8 and 10 show slightly higher Precision and AUC, they require approximately 50% of the dataset for training (nearly 20% more than the adaptive approach). By using only 31.90% of the data, the adaptive strategy attains comparable or better overall performance, demonstrating that dynamically adjusting the query batch size maximizes model performance while minimizing labeling effort, which is a key goal in active learning. Figure 7 demonstrates the Comparison of Accuracy and Training Data Percentage across Different Query Sizes.

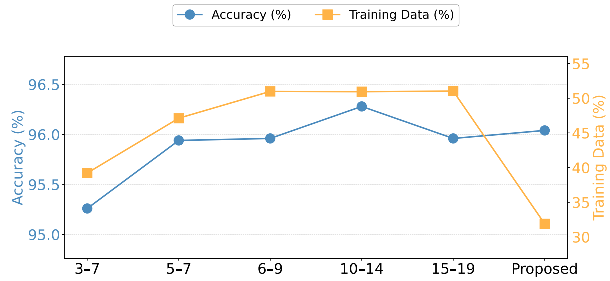

FIGURE 8. Accuracy vs. Training data for different adaptive query sizes.

F. ABLATION STUDY ON THE QUERY SIZE RANGE

We selected `list(range(3, 6))` based on an ablation study that compared several alternative query-size intervals, including `list(range(5, 8))`, `list(range(6, 10))`, `list(range(3, 8))`, `list(range(10, 15))`, and `list(range(15, 20))`. All configurations were evaluated under identical experimental conditions, and results were averaged over 10 runs. The goal of this study was to determine whether wider or shifted ranges could improve predictive performance or labeling efficiency. Table 4 reports the results in terms of Accuracy, Precision, Recall, F1-Score, AUC, and the percentage of training data used. Among the alternatives, `list(range(10, 15))` performs competitively across several metrics, showing slightly higher Accuracy (96.28%), Precision (96.75%), and AUC (98.19%). However, its Recall (96.09%) and F1-Score (96.40%) remain lower than those of the proposed configuration. More critically, it requires labeling 50.94% of the training data (about 19% more than our proposed range) which reduces its suitability for active learning, where minimizing annotation cost is crucial. Similarly, list(range(15, 20)) provides a marginal AUC improvement but requires labeling over 51% of the data, further demonstrating the inefficiency of larger query-size intervals.

In contrast, the proposed range (`list(range(3, 6))`) achieves the highest Recall (98.07%) and F1-Score (96.62%), while maintaining competitive Accuracy (96.04%), Precision (95.24%), and a strong AUC (97.08%). Importantly, it achieves this performance using only 31.90% of the training data (substantially less than any other configuration), highlighting both its predictive strength and superior labeling efficiency. Figure 8 illustrates the proportion of training data and model accuracy.

G. ABLATION STUDY ON INITIAL LABELED SET

An ablation study was conducted to evaluate the impact of varying the proportion of initially labeled samples on model performance. Table 5 reports the results for initial training sizes ranging from 1% to 12% of the dataset, along with our proposed strategy using only 1.5% of labeled data. All experiments were conducted under identical conditions, and results were averaged over 10 runs to ensure reliability.

TABLE 3. Performance for fixed and adaptive query sizes.

Query Size	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)	Training Data (%)
3	93.00	91.12	97.13	94.01	95.18	24.29
4	94.89	94.51	96.93	95.69	96.51	31.90
5	95.26	95.18	96.76	95.94	96.94	39.51
8	96.00	95.77	96.55	96.13	98.07	50.63
10	95.95	96.06	96.06	96.04	98.40	50.24
adaptive query size	**96.04**	**95.24**	**98.07**	**96.62**	**97.08**	**31.90**

TABLE 4. Performance comparison for different query size ranges.

Query Sizes Range	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)	Training Data (%)
list(range(3, 8))	95.26%	94.71%	97.42%	96.04%	96.42%	39.22%
list(range(5, 8))	95.94%	95.93%	96.91%	96.39%	97.90%	47.12%
list(range(6, 10))	95.96%	95.81%	96.30%	96.05%	98.42%	50.98%
list(range(10, 15))	96.28%	96.75%	96.09%	96.40%	98.19%	50.94%
list(range(15, 20))	95.96%	95.96%	96.09%	96%	98.53%	51.03%
Proposed list(range(3, 6))	**96.04%**	**95.24%**	**98.07%**	**96.62%**	**97.08%**	**31.90%**

As shown in the Table, the proposed 1.5% initial set achieves the best performance across all metrics, with an accuracy of 96.04%, F1-score of 96.62%, and AUC of 97.08%, while requiring only 31.90% of the total training data. In contrast, using 12% of the dataset as the initial set achieves only a marginal improvement in AUC while consuming around 11% more data initially and around 10% more of the total training data. Consequently, larger initial sets were not explored, as they would further increase labeling costs without yielding significant performance gains.

H. COMPARATIVE ANALYSIS WITH BASELINE ACTIVE LEARNING METHODS

To validate the effectiveness of the proposed framework, a comparative analysis was performed against four baseline active learning (AL) strategies: Random Sampling [34], Certain Sampling [35], Diversity Sampling [36], and Confidence-Weighted Query-by-Committee [30]. Each method was implemented under identical experimental conditions using the same heart disease dataset, initialized with 1.5% labeled data. To ensure statistical reliability and minimize the influence of random variations, all experiments were repeated over 10 runs, and the reported performance metrics represent the averaged results across these runs. As shown in Table 6, the Random Sampling achieved 91.32% accuracy, 91.50% precision, 91.52% recall, 91.47% F1-score, and 96.42% AUC using 31.90% of the labeled training data. In this baseline, samples are selected randomly from the unlabeled dataset during each active learning iteration. In Certain Sampling, instances are chosen from the unlabeled dataset in a predefined or fixed sequence, without employing any uncertainty or diversity criteria. the model recorded an accuracy of 91.53%, precision of 92.97%, recall of 90.33%, F1-score of 91.59%, and an AUC of 96.44%. An analysis of individual criteria was conducted to evaluate the importance of uncertainty and diversity. Two experiments were performed to assess the effect of each criterion separately. In the first experiment,

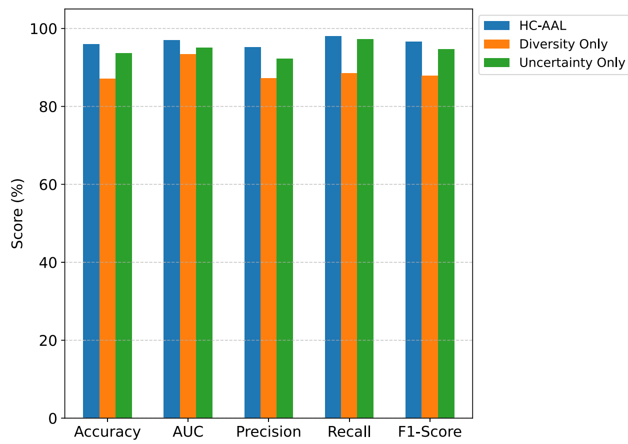

FIGURE 9. Comparison of the HC-AAL with uncertainty-only and diversity-only.

only the diversity criterion was considered. The model selects samples purely based on diversity, covering the feature space without considering model uncertainty. With Diversity Sampling, performance metrics were: 87.16% accuracy, 87.20% precision, 88.56% recall, 87.84% F1-score, and 93.46% AUC. In the second experiment, we focused on uncertainty. The model chooses samples solely based on the uncertainty estimated by the committee scores. The model achieved 93.67% accuracy, 92.24% precision, 97.27% recall, 94.67% F1-score, and 95.06% AUC. When integrating both uncertainty and diversity criteria, the proposed model achieves over 96% accuracy, 95% precision, 98% recall, a 96% F1-score, and a 97% AUC. A comparison of the proposed method with the two single-criterion experiments is presented in Fig. 9, and its performance relative to all baseline methods is shown in Fig. 10.

I. COMPARATIVE ANALYSIS WITH EXISTING APPROACHES

This section presents a comparison of the proposed method with existing approaches for heart disease classification,

TABLE 5. Performance metrics for different initial training sizes.

InitialTrain Size	Accuracy	Precision	Recall	F1-Score	AUC	Training % of Total Data
1%	95.21	94.03	97.96	95.95	96.07	31.41
2%	95.41	94.63	97.65	96.11	96.04	32.39
3%	94.99	93.67	98.04	95.79	96.23	33.37
3.5%	95.50	94.43	98.01	96.18	96.18	33.85
5%	95.16	94.57	97.31	95.90	96.26	35.41
8%	95.30	94.71	97.51	96.08	96.93	38.44
12%	95.84	95.54	97.46	96.49	97.68	42.44
Proposed	**96.04**	**95.24**	**98.07**	**96.62**	**97.08**	**31.90**

TABLE 6. Comparative analysis with baseline active learning methods.

Model	Training Data (%)	Accuracy (%)	Precision(%)	Recall (%)	F1-Score(%)	AUC(%)
Random Sampling	31.90	91.32	91.50	91.52	91.47	96.42
Certain Sampling	31.90	91.53	92.97	90.33	91.59	96.44
Diversity Sampling	31.90	87.16	87.20	88.56	87.84	93.46
Confidence-Weighted Query-by-Committee	31.90	93.67	92.24	97.27	94.67	95.06
Proposed model	**31.90**	**96.04**	**95.24**	**98.07**	**96.62**	**97.08**

TABLE 7. Comparison of the proposed method with existing approaches for heart disease classification.

Reference	Training Data	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
Ref [5]	70%	DNN + PCA + LR	93.33	95	90.48	92.68	93.7
Ref [17]	80%	SVM	91.67	92.31	88.89	90.56	Not calculated
Ref [18]	80%	HRFLM	92.95	93.08	92.88	92.47	Not calculated
Ref [19]	80%	GMERF	75	Not calculated	75	Not calculated	80
Ref [20]	90%	Optimized XGBoost	95.45	100	92.86	96.3	94.4
Ref [21]	80%	XGBoost	95.24	98.11	92.54	95.43	98.00
Ref [24]	80%	CART	87.25	88.24	84.51	Not calculated	Not calculated
Ref [25]	70%	Catboost	90.94	90	91	92.3	92.7
Ref [4]	70%	MPKNN-AL	92.22	71.42	93.75	81.08	Not calculated
Proposed model	**31.90%**	**HC-AAL**	**96.04**	**95.24**	**98.07**	**96.62**	**97.08**

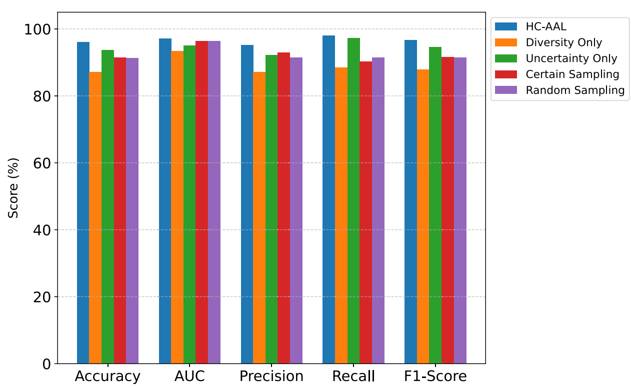

FIGURE 10. Comparison of the HC-AAL with baselines.

as summarized in Table 7, The proposed model (HC-AAL) achieves superior performance compared with recent state-of-the-art methods for heart disease classification. The model in [5], trained on 70% of the dataset, reached an accuracy of 93.33% with a precision of 95% and a recall of 90.48%. The SVM model in [17], trained on 80% of the data, reached 91.67% accuracy and 92.31% precision. Similarly, the HRFLM model in [18] also utilized 80% of the training data and obtained an accuracy of 92.95%. The GMERF method in [19], which was trained on 80% of the dataset, showed a relatively low performance with an accuracy of 75% and an AUC-ROC of 80%. The Optimized XGBoost model in [20] used 90% of the training data to achieve a higher accuracy of 95.45%, along with 100% precision and 92.86% recall. The standard XGBoost model in [21], trained on 80% of the data, reported 95.24% accuracy, 98.11% precision, and an AUC-ROC of 98%. the CART model in [24] used 80% of the data and achieved 87.25% accuracy, whereas the CatBoost approach in [25] trained on 70% of the dataset and reached 90.94% accuracy. Reference [4], a probabilistic KNN with MCMC-based metric learning and active learning, trained on 70% dataset. It achieved 92.22% accuracy and 81.08% F1-score. In contrast, Active learning (AL) aims to achieve high accuracy with minimal labeled data by selectively querying the most informative samples [27]. In the proposed HC-AAL framework, this is accomplished through a combination of uncertainty-based and diversity-based sampling. As shown, with only 31.9% of the data labeled, the model achieves a high accuracy of 96.04%, 98.07% recall, 96.62% F1-score, and 97.08% AUC-ROC, highlighting its strong performance and data efficiency while significantly reducing labeling cost.

V. CONCLUSION AND FUTURE WORK

This study introduced a heterogeneous committee-based adaptive active learning (HC-AAL) framework for efficient and accurate heart disease prediction. By integrating a diverse committee of machine learning models with a CW-QBC uncertainty measure, diversity-enhancing K-Means sampling, and an adaptive query-length strategy, the proposed method effectively reduces dependence on large labeled datasets while maintaining high predictive performance.

A comprehensive comparison with existing state-of-the-art models further demonstrates the effectiveness of the proposed framework. Traditional approaches, such as SVM, CART, CatBoost, HRFLM, GMERF, and optimized XGBoost, typically rely on 70–90% of labeled data to reach accuracies ranging between 75% and 95.45%. In contrast, the HC-AAL model achieves 96.04% accuracy, 98.07% recall, 96.62% F1-score, and 97.08% AUC-ROC while utilizing only 31.9% of labeled samples. This clearly highlights its superior data efficiency and strong predictive capability.

Overall, the HC-AAL framework demonstrates that combining adaptive active learning with heterogeneous committee learning can substantially reduce labeling costs without compromising model performance. These findings provide a promising pathway for developing scalable, cost-effective diagnostic systems in medical domains where expert annotation is limited.

Future work includes extending HC-AAL to multi-modal data, integrating semi- or self-supervised learning to reduce labeling further, developing real-time adaptive query strategies, and validating the approach on larger, more diverse datasets to enhance generalizability.

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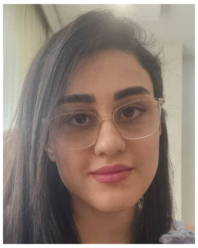

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